



ST BENEDICT'S COLLEGE

SUBJECT	LIFE SCIENCES	DATE	9 FEBRUARY 2022
GRADE	11	MARKS	50
EXAMINER	MRS S. STEGMANN	MODERATOR	MRS B. ZAJAC
NAME		DURATION	1 HOUR
CLASS			

INSTRUCTIONS:

1. This paper consists of 6 pages. Please check that your question paper is complete.
 2. Answer all questions on the folio paper provided.
 3. Write in blue or black pen only.
 4. Read all questions carefully and answer in as much detail as possible.
 5. All diagrams should have a clear heading.
 6. It is in your own interest to write legibly and to present your work neatly.
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QUESTION	POSSIBLE MARK	MARK ACHIEVED
1	22	
2	28	
TOTAL: 50		

Question 1:

- 1.1. Various possibilities are suggested as answers to the following questions. Indicate the correct answer by writing the correct letter from the corresponding question on the folio paper provided. **(5)**

1.1.1. Which of the following receives blood from the efferent arteriole?

- A. The renal vein
- B. The glomerulus
- C. The afferent arteriole
- D. The peritubular capillaries

1.1.2. Which of the following factors could possibly result in a person excreting more urine than normal?

- A. A high reabsorption of water in the proximal tubules
- B. A low concentration of ADH in the blood
- C. Low blood pressure because of too little tissue fluid.
- D. Excessive intake of salty foods.

1.1.3. Which of the following is least likely to be present in the glomerular filtrate (the filtrate produced by the nephron before it enters the loop of Henle) of a healthy adult nephron?

- A. Amino acids
- B. Glucose
- C. Electrolytes
- D. Proteins

1.1.4. The renal artery is a division of which main blood vessel?

- A. The pulmonary artery
- B. Inferior vena cava
- C. Aorta
- D. Superior vena cava

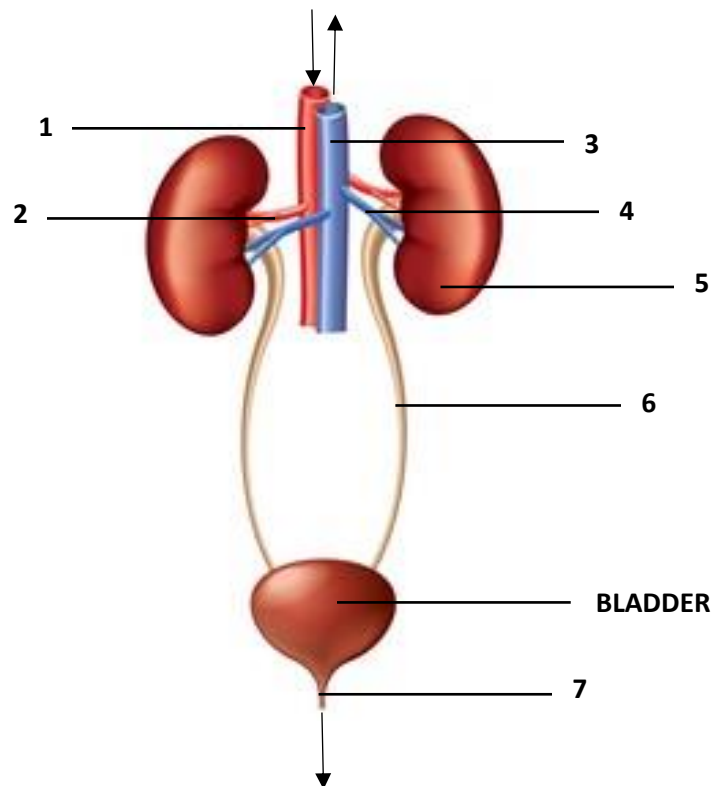
1.1.5. Which of the following does the renal system NOT play a direct role in?

- A. Regulation of blood solute concentrations
- B. Blood temperature regulation
- C. Blood pressure regulation
- D. Blood pH regulation

- 1.2. Indicate whether each of the statements in COLUMN I applies to A only, B only, both A and B or none of the items in COLUMN II. Write A only, B only, both A and B or none next to the question number on your answer sheet. (4)

COLUMN I	COLUMN II
1.2.1. Nitrogenous waste product	A: Uric acid B: Urea
1.2.2. Removal of urine from the bladder	A: Secretion B: Excretion
1.2.3. Secreted by the adrenal gland	A: Adrenalin B: ADH
1.2.4. Urine composition	A: Glucose B: Amino acids

- 1.3. The diagram below shows the excretory system of a human being. Study the diagram and answer the questions that follow.



- 1.3.1. Provide labels for structures 1, 4, 6 and 7. (4)
- 1.3.2. What is the value of the kidneys being surrounded by a thick layer of adipose tissue? (2)

1.3.3. Tabulate two differences in the blood found in structures 2 and 4. (4)

1.3.4. Explain why:

A. Urine is slightly thicker in summer than in winter. (2)

B. The renal cortex has a dotted appearance. (1)

(13)

Question 2:

2.1. Study the following table that shows the flow rate and concentration of certain substances taken at regions A, B, C and D of the nephron in the human kidney.

Part of nephron	Flow rate (cm ³ / min)	Solute concentrations (g/100 cm ³)				
		Proteins	Glucose	Sodium ions	Ammonium ions	Urea
A	4	0	0	0,6	0,04	1,80
B	200	0	0,10	0,72	0	0,05
C	40	0	0	0,3	0	0,15
D	2000	7	0,10	0,72	0	0,05

2.1.1. State and provide a reason which of the parts (A, B, C or D) of the nephron represent the following:

a. Afferent arteriole (2)

b. Bowman's capsule (2)

c. Loop of Henle (2)

d. Collecting duct (2)

2.1.2. State two functions of the kidney that can be supported by the data given in the table. (2)

2.1.3. Provide a labeled diagram of the renal corpuscle. (5)

(15)

2.2. Study the information below and answer the questions that follow.



Combur-10 Test Strip

Streptococcus bacteria can cause serious infection of different parts of the urinary system.

The Combur-10 Test Strip is a dry chemistry test for the early and reliable detection of kidney diseases, diabetes, and urinary tract infection, like cystitis.



Dip the test strip in urine for about 1 second.



Wipe the edge against the rim of the vessel to remove excess urine when withdrawing the strip.



After 60 seconds, compare the colour of the detection pad of the strip with the colour scale on the test strip vial. Strips face downwards during operation and fingers stay clear and safe from the samples.

- 2.2.1. A urinary tract infection (UTI) is caused by bacteria entering the urinary system. Which part of the urinary system does the bacteria most likely enter? (1)
- 2.2.2. Why is early detection, provided by a test kit like Combur-10 Test Strip, so vital? (3)
- 2.2.3. Name the UTI for which Combur-10 Test Strip is very commonly used to test for. (1)

- 2.2.4. A study was done to determine if UTIs were more common amongst young males or females.

The results of a test group of individuals suffering with UTIs at any given time is tabulated below.

Age group (years)	Female percentage affected (%)	Male percentage affected (%)
<1	39.1	60.9
1-5	73.2	26.8
6-10	82.4	17.6
11-15	66.7	33.3
Total	70.0	30.0

- A. Provide TWO variables that needed to be controlled in the above investigation. (2)
- B. Provide a conclusion for the above study. (1)
- C. Display the percentage of males affected in the different age groups in an appropriate graph. (5)
- (13)**

TOTAL: 50 MARKS